Lindsey Ma

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Education

Columbia University, M.S. in Statistics, GPA 3.71 *Sep 2024 – May 2026*
Coursework: Deep Learning, Advanced Machine Learning, Robot Learning, Statistical Analysis of Neural Data
Renmin University of China, B.A. in Economics, GPA 3.82 *Sep 2019 – Jun 2023*
Coursework: Linear Algebra, Calculus, Probability Theory

Experience

Research Engineer Intern – Applied ML for Plant Identification *May 2026 – Present*
Arborphy AI LLC *New York, NY*

- Built a plant species classifier on ∼9,000 images spanning 200+ species, designing an EfficientNet–BioCLIP dual-encoder with bidirectional spatial cross-attention, auxiliary organ supervision, and BioCLIP-anchored gated residuals; improved fusion macro-F1 from 0.808 to 0.832 across architecture iterations.
- Designed a leakage-aware evaluation protocol (individual-disjoint splits, multi-seed runs, grouped held-out sets) to benchmark single encoders, feature concatenation, and uni- vs. bidirectional cross-attention; showed that fusion gains did not reliably exceed a BioCLIP-only baseline.
- Benchmarked frozen BioCLIP, DINOv3, and EfficientNet embeddings with linear probes and cross-species retrieval; DINOv3 reached 87.3% organ-classification accuracy and 84.3% retrieval P@1.
- Analyzed what each backbone encodes using multi-seed clustering, nearest-neighbor retrieval, local PCA rotations, and centroid shrinkage, separating species identity and organ morphology from photographic-scale confounds.
- Built a hierarchical multi-agent reasoning prototype in which a top-level agent can delegate to specialized visual-analysis agents or generate new tools on demand, with constrained iterative planning, per-step information-gain scoring, and knowledge-graph-based evidence aggregation.

Research Assistant (ML) – Self-Supervised Denoising for Neural Imaging *Jan 2026 – May 2026*
Center for Theoretical Neuroscience, Columbia Zuckerman Institute *New York, NY*

- Designed a four-direction spatial masking strategy for self-supervised denoising of two-photon calcium imaging, integrated with PMD spatial-component selection; reduced residual energy by up to 20%.
- Trained and compared ResNet, U-Net, and lightweight CNN denoisers; ran systematic ablations over receptive field, normalization, loss function, training settings, and spatial-component selection.
- Studied Noise2Noise training-pair construction (shared vs. independent temporal bases, frame fractions, independent corruptions) to characterize component stability and limits of self-supervision.
- Built synthetic benchmark datasets from real neuronal spatial footprints, simulated temporal traces, and controlled Gaussian noise to measure reconstruction quality and signal recovery across noise levels.
- Evaluated spatial, temporal, and combined denoising at component and reconstructed-video levels, quantifying trade-offs in signal retention, temporal correlation, reconstruction fidelity, and rank selection.
- Diagnosed a numerical discrepancy in PMD's blockwise spatial update and contributed a fixed-temporal-basis least-squares refitting fix to the codebase through pull requests.
- Packaged lab research code into a downloadable software package, with public documentation and step-by-step guides for wet-lab researchers.

Software Developer – Real-Time Vision for Robotics *May 2025 – Dec 2025*
InnoX Academy Incubator Robotics Lab *Shenzhen, China (Remote from Sep 2025)*

- Built a real-time vision pipeline for an autonomous badminton-training robot, integrating YOLOv8 player detection, pose estimation, shuttlecock tracking, and court localization at 20+ FPS.
- Implemented stereo 3D shuttlecock tracking and landing-point prediction.
- Deployed models on Raspberry Pi with ONNX Runtime, quantization, and an asynchronous video pipeline, reducing inference latency by 8%.
- Integrated a locally hosted language model (Ollama) to map voice commands to robot control actions.

Selected Projects & Activities

Hierarchical Multi-Agent System with Audit-Driven Trust *Oct 2026 – Present*

- Architected a coordinator–manager–worker agent hierarchy with independent verifier agents per pipeline, deployed in TypeScript as a scheduled daily service on a self-hosted VPS.
- Designed per-stage trust levels that gate each agent's permissions (human-confirmed vs. autonomous): a verifier on a different model blind-audits sampled outputs, stages auto-demote above 10% error, and only a human can promote them.
- Turned every audit disagreement into a rule or regression test, cutting a worker's false-positive rate from 27.0% to 6.1%; kept deterministic checks in code and logged irreversible actions in an append-only ledger.

Hackathon Team Lead – 2nd Place, XRPL ($5K); 3rd Place, Stellar ($3K) *Sep 2025 – Oct 2025*

- Led technical development for two award-winning 4-person teams, owning product architecture, implementation, integration, and debugging, and coordinating team strategy and judge presentations. [Code]
- Built VenMorph, an XRPL payment platform with wallet integration, Flare price oracles, payment requests, and on-chain transaction verification; and a Stellar structured-finance platform with senior/junior tranches, cash-flow waterfalls, loss simulation, and Soroban smart contracts.

LLM Interpretability – Technical Fellow, Columbia AI Alignment Club *Feb 2025 – May 2025*

- Reproduced Anthropic's circuit-tracing analysis of hallucination on Qwen3-4B using pretrained transcoders and attribution graphs, examining candidate refusal mechanisms and limits of circuit faithfulness.
- Built interpretability tooling across all 28 transformer layers (activation hooks, linear probes, bidirectional steering) with randomized-direction, sham-intervention, and grouped-bootstrap controls; verified Hugging Face–TransformerLens consistency for Qwen3-8B FP32 inference (logit discrepancy $< 4.2 \times 10^{-5}$).
- Ran controlled known- vs. unknown-entity experiments on Qwen3-8B (40 responses, matched prompt templates, multiple seeds), finding answer withholding too unstable for reliable causal circuit claims.
- Traced token-level generation trajectories and identified a case where factual attribution and answer withholding diverged at the 7th generated token, showing that first-token logit shifts alone cannot establish semantic hallucination effects.

Bayesian Modeling of the FlyWire Connectome – Computational Statistics *May 2025*

- Implemented Bayesian stochastic block models with class-dependent and spatially informed likelihoods to infer latent neuron types from FlyWire chemical-synapse networks; evaluated cell-type recovery across brain regions and identified where distance-dependent assumptions break down.
- Built a burst-based Poisson spike-train simulator and a joint structural–functional inference objective using co-firing connectivity. [Paper]

Real-Time EEG Motor-Imagery Decoding – Brain-Computer Interface Lab *Jan 2025*

- Built an EEG decoding pipeline with CSP features, training-only fitting, and run-grouped validation, raising offline accuracy from 67.90% to 77.39% over a CSP–LDA baseline.
- Implemented streaming inference with stateful causal filtering, ring buffers, and cue-triggered prediction; achieved 77.78% cross-day accuracy and exact streaming–batch agreement across 1,296 predictions.

Data Analyst – Columbia Asian Faculty Association *Jan 2025 – Mar 2025*

- Independently carried out end-to-end analysis of a university-wide survey of Asian-heritage faculty under faculty supervision, covering well-being, workload, caregiving, institutional support, and career concerns; cleaned and modeled responses in R and Python with subgroup comparisons and regression analysis.
- Authored a formal statistical report and presented findings to the association's board across multiple briefings, informing board-level decisions on faculty support.

Technical Skills

Languages:	Python, SQL, R
ML & Deep Learning:	PyTorch, Hugging Face Transformers, scikit-learn, TransformerLens
Inference & Deployment:	ONNX Runtime, model quantization, vLLM, Ollama, FastAPI
Infrastructure & Tools:	Linux, VPS self-hosting (bots, autonomous agents), Docker, Git, Neo4j